

Basic Algebra

Course Notes MA2205: Basic Algebra

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Introduction

Around the year 1800, people already knew how to solve equations of the forms -

- Linear Equations: $ax + b = 0$
- Quadratic Equations: $ax^2 + bx + c = 0$
- Cubic Equations: $ax^3 + bx^2 + cx + d = 0$
- Quartic Equations: $ax^4 + bx^3 + cx^2 + dx + e = 0$

However, the general solution for equations of degree 5 or higher was not known. A young mathematician named Evariste Galois answered the question of the existence of a general solution for polynomial equations of degree 5 or higher, using a tool called groups.

Around the same time, Carl Friedrich Gauss introduced a technique called modular arithmetic, which laid the foundation for number theory. This technique shares a close relationship with groups.

The 1800s also saw a revolution in geometry. For over 2000 years, Euclidean Geometry was the only known geometry. However, mathematicians began to explore different geometries (specially geometries on non-planar or curved surfaces). Groups became useful in studying these new geometries as well.

The formalization of groups by mathematicians supporting Galois and Gauss led to the birth of Group Theory and Abstract Algebra. Mathematicians realized that groups were very useful in studying various mathematical structures, and thus extended the idea of groups to other structures such as rings, fields, modules, vector spaces etc. It took years of effort by many mathematicians to formalize these structures and study their properties.

As time passed, the applications of these structures in various fields of mathematics and science became apparent. Today, it is used in computer science, cryptography, physics, chemistry, biology and many other fields.

This course aims to introduce the basic concepts of Group Theory, Ring Theory and some fundamental ideas in Abstract Algebra. We will explore the properties of these structures and their applications in various fields of mathematics and science.

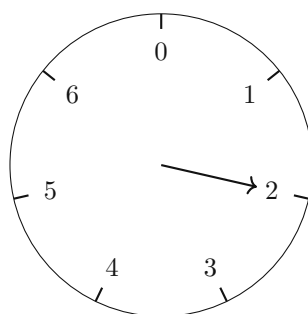
Chapter 1

Groups

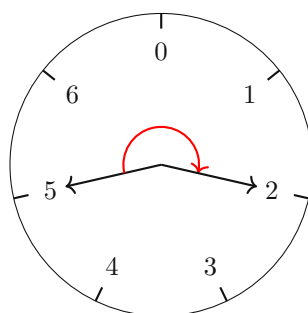
Before we formally define groups, we consider a few examples to get a basic idea of what groups are and how they arise in various mathematical contexts.

1.1 Examples of Groups

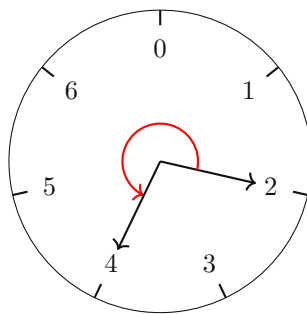
1. **Clock Arithmetic:** Consider a planet which spins faster than the Earth, such that a day on this planet is 7 hours long. The time on this planet can be represented using numbers from 0 to 6, where 0 represents midnight, 1 represents 1 AM, and so on. Let us visualise this using a clock with 7 hours.



Let us define an operation of addition on this clock, where adding a number a to the current time b results in moving a hours forward on the clock. For example, if the current time is 5 and we add 4 hours, we move 4 hours forward on the clock, resulting in a time of 2 (since we wrap around after reaching 6). Let us visualise this operation on the clock.



Here, we see that $5 + 4 = 2$. Similarly, we can define subtraction on this clock, where subtracting a number a from the current time b results in moving a hours backward on the clock. For example, if the current time is 2 and we subtract 5 hours, we move 5 hours backward on the clock, resulting in a time of 4 (since we wrap around after reaching 0). Let us visualise this operation on the clock.

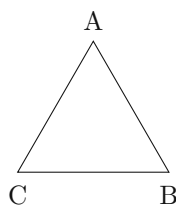


Notice that $2 - 5 = 4$, and also that subtraction can be thought of as adding the negative of a number, i.e., $2 - 5 = 2 + (-5) = 2 + 2 = 4$.

This type of arithmetic is called modular arithmetic, specifically arithmetic modulo 7 in this case. The set of numbers $\{0, 1, 2, 3, 4, 5, 6\}$ with the operations of addition and subtraction defined as above forms a mathematical structure known as a group. In this group, the operation of addition is closed, associative, has an identity element (0), and every element has an inverse (for example, the inverse of 5 is 2 since $5 + 2 \equiv 0 \pmod{7}$).

Modular arithmetic has applications in various fields such as cryptography, computer science, and number theory.

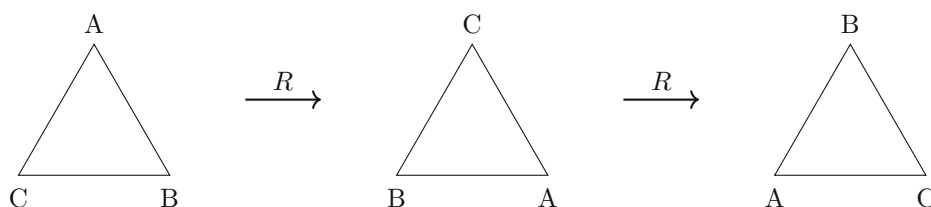
2. **Symmetries of an Equilateral Triangle:** Consider an equilateral triangle with vertices labeled A, B, C . The symmetries of this triangle include rotations and reflections that map the triangle onto itself. Let us visualize these symmetries.



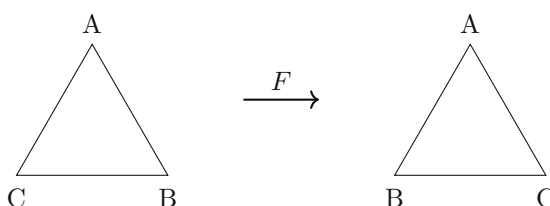
The operations that map the triangle onto itself are:

- **Rotations:** We can rotate the triangle about its center by 120° clockwise, to get the triangle mapped onto itself. [Denoted by R .]
- **Flipping:** Flipping the triangle about an axis passing through a vertex and the midpoint of the opposite side results in mapping the triangle onto itself. [Denoted by F .]
- **No Operation:** The triangle remains unchanged [Denoted by I].

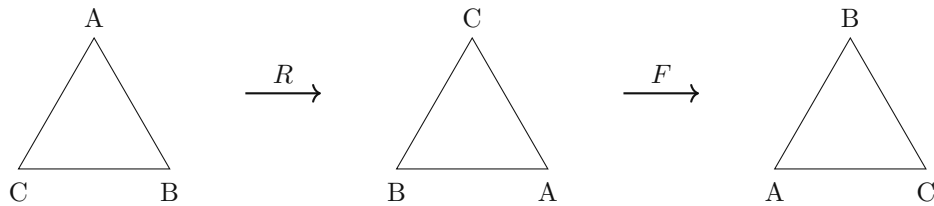
We can compose these fundamental operations to get all the symmetries of the triangle. Let us visualise these operations.



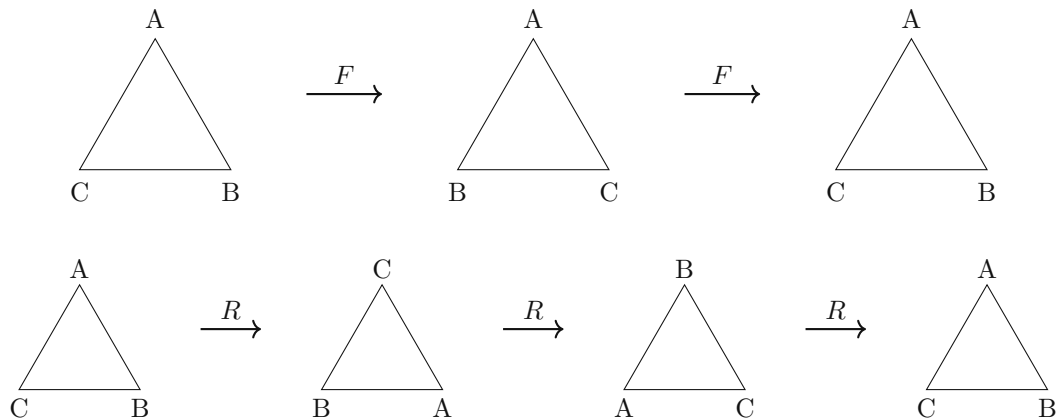
Similarly, we can visualize the flipping operation.



Let us now compose the flipping operation with rotations to get more symmetries.



Continuing this process, we find that there are a total of 6 symmetries of the equilateral triangle: I, R, R^2, F, FR, FR^2 . Note that double flipping results in the identity operation, i.e., $F^2 = I$ and that $R^3 = I$. Let us visualise the above two operations to verify that they result in the identity operation.



3. **Integers under Addition:** Consider the set of integers $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ with the operation of addition.

Note: We can define subtraction as adding the negative of a number, i.e., $x - y = x + (-y)$. We now have the following properties of this set equipped with the addition operation -

- Closure: Let $x, y \in \mathbb{Z}$, then $x + y \in \mathbb{Z}$.
- Additive Identity: There is a unique number $0 \in \mathbb{Z}$ such that for any $x \in \mathbb{Z}$, $x + 0 = x$. This number is called the identity element.

Exercise 1.1: Instead of an equilateral triangle, consider the following shapes and find all their symmetries -

- A square
- A rectangle (not a square)
- A cube
- A Tetrahedron

Soln: Let us consider just the case of a square here.

Square: Consider a square with vertices labeled A, B, C, D . We can do the operations of rotation by 90° clockwise (denoted by R), Flipping about the vertical axis (F_v), flipping about diagonal axis (F_d) and no operation (I). We can compose these operations to get all the symmetries of the square. There are a total of 8 symmetries of the square.

Similarly, we can find the symmetries of the other shapes.

Let us now analyze the similarities between these examples.

1. Each of these examples has a set of objects that are considered (numbers of the clock, symmetries of the triangle, integers). We call these objects elements of a set.

2. We have a way of combining two elements of the set to get another element of the set (addition on the clock, composition of symmetries, addition of integers). We call this way of combining elements an operation.
3. All sets are closed under the operation defined on them (adding two numbers on the clock gives another number on the clock, composing two symmetries gives another symmetry, adding two integers gives another integer).
4. The operation is associative (for any three numbers on the clock, adding them in any order gives the same result, composing three symmetries in any order gives the same result, adding three integers in any order gives the same result).
5. There is an identity element in each set with respect to the operation (0 on the clock, identity symmetry of the triangle, 0 in integers).
6. Every element has an inverse with respect to the operation (for any number on the clock, there is another number that when added gives 0, for any symmetry of the triangle, there is another symmetry that when composed gives the identity symmetry, for any integer, there is another integer that when added gives 0).

1.2 Groups and Operations

1.2.1 Definitions

Based on the observations from the previous section, we can now formally define groups and operations.

Definition 1.2.1.1

Groups

A group $(G, *)$ is a set G equipped with a **binary operation** $* : G \times G \rightarrow G$ that satisfies the following properties:

- **Closure:** For all $a, b \in G$, the result of the operation $a * b$ is also in G .
- **Associativity:** For all $a, b, c \in G$, $(a * b) * c = a * (b * c)$.
- **Identity Element:** There exists an element $e \in G$ such that for every element $a \in G$, $e * a = a * e = a$.
- **Inverse Element:** For each $a \in G$, there exists an element $b \in G$ such that $a * b = b * a = e$, where e is the identity element.

It took years of effort to develop the formal definition of groups based on the most fundamental properties observed in various mathematical contexts. For example, the property of commutativity (i.e., $a * b = b * a$ for all $a, b \in G$) is not included in the definition of groups, as there exist many important groups where this property does not hold. Groups where commutativity holds are called abelian groups, named after the mathematician Niels Henrik Abel. Similarly, a non-commutative group is called non-abelian.

Let us now formally define a binary operation.

Definition 1.2.1.2

Binary Operation

A binary operation on a set S is a function $* : S \times S \rightarrow S$ that combines any two elements $a, b \in S$ to produce another element $c \in S$, denoted as $a * b = c$.

Let us consider some examples of binary operations.

1. **Addition on Integers:** The operation of addition $+ : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $(a, b) \mapsto a + b$ is a binary operation on the set of integers $\mathbb{Z}$.

2. **Composition of Symmetries:** Consider a set X equipped with some structure. Now, a symmetry of X is a bijective function from X to itself that preserves the structure of X . The operation of composition of symmetries $\circ : \text{Sym}(X) \times \text{Sym}(X) \rightarrow \text{Sym}(X)$ defined by $(f, g) \mapsto f \circ g$ is a binary operation on the set of symmetries $\text{Sym}(X)$.
3. **Union and Intersection:** Let S be a non-empty set. Then, $\cup$ and $\cap$ are binary operations on the power set of S , denoted by $\mathcal{P}(S)$. We define the operations as follows:

$$\cup : \mathcal{P}(S) \times \mathcal{P}(S) \rightarrow \mathcal{P}(S)$$

defined by $(A, B) \mapsto A \cup B$

$$\cap : \mathcal{P}(S) \times \mathcal{P}(S) \rightarrow \mathcal{P}(S)$$

defined by $(A, B) \mapsto A \cap B$

4. **Matrix Multiplication:** For $n \in \mathbb{N}, n \geq 1$, let $GL_n(\mathbb{R})$ be the set of all invertible $n \times n$ matrices with real entries. Then, the operation of matrix multiplication $\cdot : GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) \rightarrow GL_n(\mathbb{R})$ defined by $(A, B) \mapsto A \cdot B$ is a binary operation on the set of invertible matrices $GL_n(\mathbb{R})$.

Exercise 1.2: Is the matrix multiplication operation associative and commutative on the set $GL_n(\mathbb{R})$?

Soln: The operation of matrix multiplication is associative but not commutative. That is, for any three matrices $A, B, C \in GL_n(\mathbb{R})$, we have $(A \cdot B) \cdot C = A \cdot (B \cdot C)$, but in general, $A \cdot B \neq B \cdot A$.

Exercise 1.3: which of the following are binary operations on $\mathbb{N}$?

1. $(a, b) \mapsto a - b$
2. $(a, b) \mapsto \pm\sqrt{ab}$
3. $(a, b) \mapsto 5$

Soln: Let us analyze each operation:

1. The operation $(a, b) \mapsto a - b$ is not a binary operation on $\mathbb{N}$ because the result of the operation may not be a natural number. For example, if $a = 3$ and $b = 5$, then $a - b = -2$, which is not in $\mathbb{N}$.
2. The operation $(a, b) \mapsto \pm\sqrt{ab}$ is not a binary operation on $\mathbb{N}$ because the result of the operation may not be a natural number. For example, if $a = 2$ and $b = 8$, then $\sqrt{ab} = \sqrt{16} = 4$, which is in $\mathbb{N}$, but the negative root -4 is not in $\mathbb{N}$.
3. The operation $(a, b) \mapsto 5$ is a binary operation on $\mathbb{N}$ because for any natural numbers a, b , the result of the operation is always 5, which is in $\mathbb{N}$.

Exercise 1.4: Is the operation on $\mathbb{R}$ defined by $(a, b) \mapsto \frac{a-b}{2}$ commutative and associative?

Soln: Let us analyze the operation $(a, b) \mapsto \frac{a-b}{2}$:

- **Commutativity:** The operation is not commutative because in general, $\frac{a-b}{2} \neq \frac{b-a}{2}$. For example, if $a = 3$ and $b = 5$, then $\frac{3-5}{2} = -1$ and $\frac{5-3}{2} = 1$, which are not equal.
- **Associativity:** The operation is not associative because in general, $\frac{(\frac{a-b}{2})-c}{2} \neq \frac{a-(\frac{b-c}{2})}{2}$. For example, if $a = 4, b = 2, c = 0$, then:

$$\frac{(\frac{4-2}{2})-0}{2} = \frac{1-0}{2} = 0.5$$

and

$$\frac{4-(\frac{2-0}{2})}{2} = \frac{4-1}{2} = 1.5$$

which are not equal.

1.2.2 Cayley Tables

Let S be a non-empty set and $*$ be a binary operation on S . We can represent the operation $*$ using a table called the Cayley table. The rows and columns of the table are labeled with the elements of S , and the entry in the row corresponding to element a and the column corresponding to element b is the result of the operation $a * b$.

Let us consider the following example -

Let $S = \{\pm 1, \pm i\}$, where i is the imaginary unit. We define a binary operation $*$ on S as follows -

$*$	1	-1	i	$-i$
1	1	-1	i	$-i$
-1	-1	1	$-i$	i
i	i	$-i$	-1	1
$-i$	$-i$	i	1	-1

Note 1.2.2.1

We note the following properties of the above Cayley table -

1. We have 1 as the identity element, since $1 * a = a * 1 = a$ for all $a \in S$.
2. Every row and every column contain the identity element, $1 \cdot 1 = 1, -1 \cdot -1 = 1, i \cdot -i = 1, -i \cdot i = 1$
3. Associativity holds, i.e., $(a * b) * c = a * (b * c)$ for all $a, b, c \in S$.
4. The table is symmetric about the main diagonal, i.e., $a * b = b * a$ for all $a, b \in S$. Thus, the operation is commutative.
5. No duplicate elements appear in any row or column. If there existed duplicates, then there would be elements without inverses.

Based on the above properties, we can conclude that the set S with the operation $*$ forms an abelian group.

We now consider a few exercises based on Cayley tables.

Exercise 1.5: Consider the group $(\{e, a, b\}, *)$. Find the cayley table for this group, given that e is the identity element, and that $a * a = b, a * b = e, b * a = e, b * b = a$. Show that the group is abelian.

Soln: We can construct the Cayley table as follows -

$*$	e	a	b
e	e	a	b
a	a	b	e
b	b	e	a

To show that the group is abelian, we need to verify that the operation is commutative, i.e., $a * b = b * a$ for all $a, b \in \{e, a, b\}$. We can see that the cayley table is symmetric about the main diagonal, which implies that the operation is commutative. Therefore, the group is abelian.

Exercise 1.6: Using Cayley tables, find all the groups of order 4 (i.e., groups with 4 elements). Hint: There are 4 possible tables, 3 are identical to each other, and 2 distinct groups exist. Soln: Let us denote the elements of the group as $\{e, a, b, c\}$, where e is the identity element. We can construct the Cayley tables for the possible groups of order 4.

Group 1:

$*$	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b	b	c	e	a
c	c	e	a	b

Group 2:

$*$	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

These are the two distinct groups of order 4. The other two possible tables are identical to one of these two groups. Thus, we have found all the groups of order 4 using Cayley tables.

Exercise 1.7: Let $(G, *)$ be a group. Prove the following -

1. The identity element is unique.
2. For each element $a \in G$, $\exists$ unique $b \in G$ such that $a * b = b * a = e$, where e is the identity element.
3. For all $a, b, c \in G$, if $a * b = a * c$, then $b = c$ (Left Cancellation Law). Similarly, if $b * a = c * a$, then $b = c$ (Right Cancellation Law).
4. For all $a, b \in G$, the equation $a * x = b$ has a unique solution for $x \in G$. Similarly, the equation $y * a = b$ has a unique solution for $y \in G$.
5. $(a^{-1})^{-1} = a$ for all $a \in G$ and $(a * b)^{-1} = b^{-1} * a^{-1}$ for all $a, b \in G$.
6. $\forall a \in G, m, n \in \mathbb{Z}, a^m * a^n = a^{m+n}$, where a^m denotes the m -th power of a with respect to the operation $*$. And, $(a^m)^n = a^{mn}$.

Soln: Let us prove each of the statements one by one -

1. Suppose there are more than 1 distinct identity elements in the group, say e_1 and e_2 . By the definition of identity element, we have $e_1 * a = a$ and $e_2 * a = a$ for all $a \in G$. In particular, for $a = e_2$, we have $e_1 * e_2 = e_2$. Similarly, for $a = e_1$, we have $e_2 * e_1 = e_1$. Thus, we have $e_1 = e_2$, which contradicts our assumption that they are distinct. Therefore, the identity element is unique.
2. Let $a \in G$ be an arbitrary element. By the definition of inverse element, there exists an element $b \in G$ such that $a * b = e$ and $b * a = e$, where e is the identity element. Suppose there exists another element $c \in G$ such that $a * c = e$ and $c * a = e$. Then, we have:

$$b = b * e = b * (a * c) = (b * a) * c = e * c = c$$

Thus, $b = c$, which shows that the inverse element is unique.

3. Let $a, b, c \in G$ such that $a * b = a * c$. We want to show that $b = c$. By the definition of inverse element, there exists an element $a^{-1} \in G$ such that $a^{-1} * a = e$, where e is the identity element. Now, we have:

$$a^{-1} * (a * b) = a^{-1} * (a * c) \implies (a^{-1} * a) * b = (a^{-1} * a) * c \implies e * b = e * c \implies b = c$$

The same logic can be applied to prove the right cancellation law.

4. We have $a, b \in G$. We want to show that the equation $a * x = b$ has a unique solution for $x \in G$. By the definition of inverse element, there exists an element $a^{-1} \in G$ such that $a^{-1} * a = e$, where e is the identity element. Now, we can multiply both sides of the equation $a * x = b$ by a^{-1} from the left:

$$a^{-1} * (a * x) = a^{-1} * b \implies (a^{-1} * a) * x = a^{-1} * b \implies e * x = a^{-1} * b \implies x = a^{-1} * b$$

Thus, the equation $a * x = b$ has a unique solution for $x \in G$. Similarly, we can show that the equation $y * a = b$ has a unique solution for $y \in G$.

5. We know that every $a \in G$ has a unique inverse element $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$, where e is the identity element. From this, we can deduce that $(a^{-1})^{-1} = a$ for all $a \in G$. Now, let $a, b \in G$. We want to show that $(a * b)^{-1} = b^{-1} * a^{-1}$. By the definition of inverse element, we have:

$$(a * b) * (b^{-1} * a^{-1}) = a * (b * b^{-1}) * a^{-1} = a * e * a^{-1} = a * a^{-1} = e$$

Similarly, we can show that $(b^{-1} * a^{-1}) * (a * b) = e$. Thus, we have shown that $(a * b)^{-1} = b^{-1} * a^{-1}$ for all $a, b \in G$.

6. We prove the properties using mathematical induction.

Base Case: For $n = 1$, we have $a^m * a^1 = a^{m+1}$ and $(a^m)^1 = a^{m \cdot 1} = a^m$, which holds true.

Inductive Step: Assume that the properties hold for some $n = k$, i.e., $a^m * a^k = a^{m+k}$ and $(a^m)^k = a^{m \cdot k}$. We need to show that the properties hold for $n = k + 1$.

We have:

$$a^m * a^{k+1} = a^m * (a^k * a) = (a^m * a^k) * a = a^{m+k} * a = a^{m+k+1}$$

and

$$(a^m)^{k+1} = (a^m)^k * a^m = a^{m \cdot k} * a^m = a^{m \cdot k + m} = a^{m \cdot (k+1)}$$

Thus, by the principle of mathematical induction, the properties hold for all $n \in \mathbb{Z}$.